2、获取距离信息（Linux）

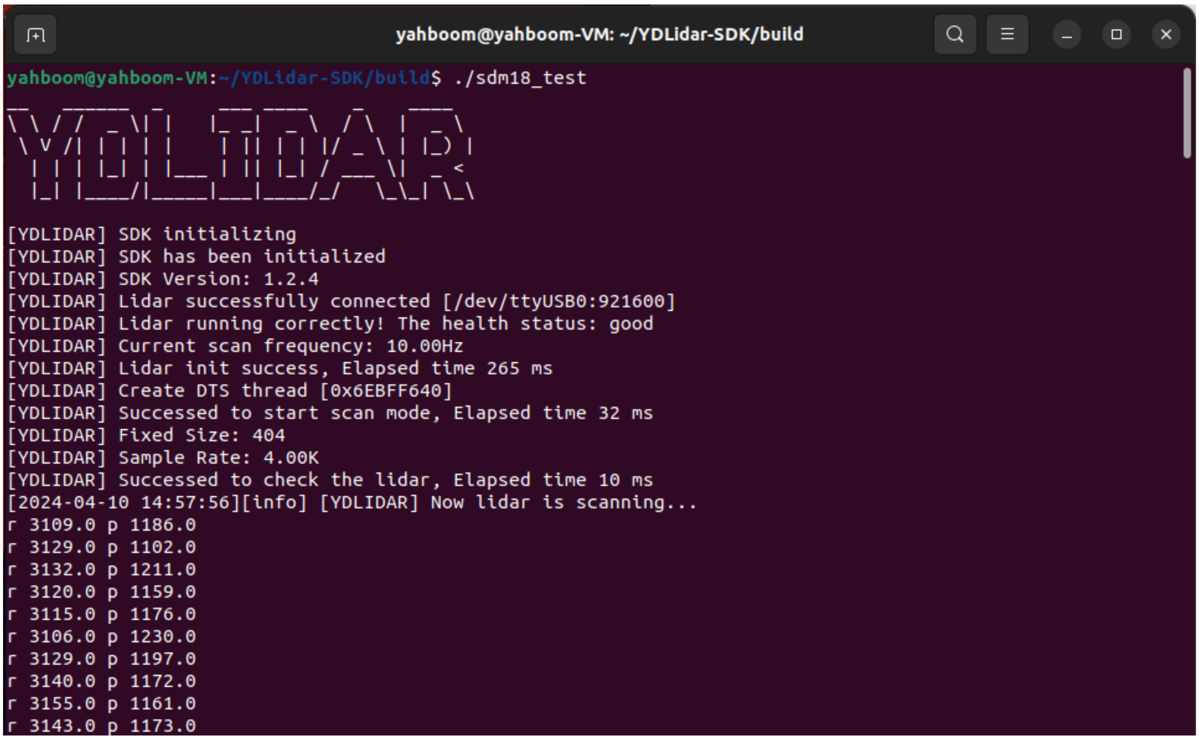
没看《使用前准备工作》的需要先看，之后再来运行下面的教程

1、启动

在编译SDK的时候，我们会生成一个sdm_test 的程序，这个程序的位置在YDLidar-SDK/build目录下，

正常编译的话是会生成这个可执行文件的，我们在这个目录下打开终端输入命令，

```
./sdm18_test
```



这里的r表示距离，p表示这个点的强度，单位是毫米mm。

2、源码

源码位置（SDK安装位置：~/YDLidar-SDK）：~/YDLidar-SDK/samples/sdm18_test.cpp

注意：根据自己的SDK安装位置进行查找，是在SDK目录下的samples里边。

```
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 *********************************************************************/
#include "CYdLidar.h"
#include <iostream>
#include <string>
#include <algorithm>
#include <cctype>
using namespace std;
using namespace ydlidar;

#if defined(_MSC_VER)
#pragma comment(lib, "ydlidar_sdk.lib")
#endif

/**
 * @brief sdm test
 * @param argc
 * @param argv
 * @return
 * @par Flow chart
 * Step1: instance CYdLidar.\n
 * Step2: set paramters.\n
 * Step3: initialize SDK and LiDAR.(::CYdLidar::initialize)\n
 * Step4: Start the device scanning routine which runs on a separate thread and
enable motor.(::CYdLidar::turnOn)\n
 * Step5: Get the LiDAR Scan Data.(::CYdLidar::doProcessSimple)\n
 * Step6: Stop the device scanning thread and disable motor.
(::CYdLidar::turnOff)\n
 * Step7: Uninitialize the SDK and Disconnect the LiDAR.
(::CYdLidar::disconnecting)\n
 */

int main(int argc, char *argv[])
{
  printf("__   ______  _     ___ ____    _    ____  \n");
  printf("\\ \\ / /  _ \\| |   |_ _|  _ \\  / \\  |  _ \\ \n");
  printf(" \\ v /| | | | |    | || | | |/ _ \\ | |_) | \n");
  printf("  | | | |_| | |___ | || |_| / ___ \\|  _ <  \n");
  printf("  |_| |____/|_____|___|____/_/   \\_\\_| \\_\\ \n");
```

```
printf("\n");
fflush(stdout);

//初始化
ydlidar::os_init();
//初始化串口号
std::string port = "/dev/ttyUSB0";

int baudrate = 921600; //默认串口号

bool isSingleChannel = false;

CYdLidar laser;
//////////////////////string property/////////////////
/// lidar port
laser.setlidaropt(LidarPropSerialPort, port.c_str(), port.size());
/// ignore array
std::string ignore_array;
ignore_array.clear();
laser.setlidaropt(LidarPropIgnoreArray, ignore_array.c_str(),
                  ignore_array.size());
//////////////////////int property/////////////////
/// lidar baudrate
laser.setlidaropt(LidarPropSerialBaudrate, &baudrate, sizeof(int));
/// sdm lidar
int optval = TYPE_SDM18;
laser.setlidaropt(LidarPropLidarType, &optval, sizeof(int));
/// device type
optval = YDLIDAR_TYPE_SERIAL;
laser.setlidaropt(LidarPropDeviceType, &optval, sizeof(int));
/// sample rate
optval = 4;
laser.setlidaropt(LidarPropSampleRate, &optval, sizeof(int));
/// abnormal count
optval = 3;
laser.setlidaropt(LidarPropAbnormalCheckCount, &optval, sizeof(int));
/// Intenstiy bit count
optval = 8;
laser.setlidaropt(LidarPropIntenstiyBit, &optval, sizeof(int));

//////////////////////bool property/////////////////
/// fixed angle resolution
bool b_optvalue = false;
laser.setlidaropt(LidarPropFixedResolution, &b_optvalue, sizeof(bool));
/// rotate 180
laser.setlidaropt(LidarPropReversion, &b_optvalue, sizeof(bool));
/// Counterclockwise
laser.setlidaropt(LidarPropInverted, &b_optvalue, sizeof(bool));
b_optvalue = true;
laser.setlidaropt(LidarPropAutoReconnect, &b_optvalue, sizeof(bool));
/// one-way communication
laser.setlidaropt(LidarPropSingleChannel, &isSingleChannel, sizeof(bool));
/// intensity
b_optvalue = true;
laser.setlidaropt(LidarPropIntenstiy, &b_optvalue, sizeof(bool));
/// Motor DTR
```

```cpp
b_optvalue = true;
laser.setlidaropt(LidarPropSupportMotorDtrCtrl, &b_optvalue, sizeof(bool));
/// HeartBeat
b_optvalue = false;
laser.setlidaropt(LidarPropSupportHeartBeat, &b_optvalue, sizeof(bool));

//////////////////////float property/////////////////
/// unit: °
float f_optvalue = 180.0f;
laser.setlidaropt(LidarPropMaxAngle, &f_optvalue, sizeof(float));
f_optvalue = -180.0f;
laser.setlidaropt(LidarPropMinAngle, &f_optvalue, sizeof(float));
/// unit: m
f_optvalue = 20.f;
laser.setlidaropt(LidarPropMaxRange, &f_optvalue, sizeof(float));
f_optvalue = 0.025f;
laser.setlidaropt(LidarPropMinRange, &f_optvalue, sizeof(float));
/// unit: Hz
float frequency = 10.0;
laser.setlidaropt(LidarPropScanFrequency, &frequency, sizeof(float));

// laser.setEnableDebug(true); //打印串口原始数据

// 雷达初始化
bool ret = laser.initialize();
if (!ret)
{
  fprintf(stderr, "[YDLIDAR] Fail to initialize %s\n", laser.DescribeError());
  return -1;
}
// 启动扫描
ret = laser.turnOn();
if (!ret)
{
  fprintf(stderr, "[YDLIDAR] Fail to turn on %s\n", laser.DescribeError());
  return -1;
}

LaserScan scan;
while (ret && ydlidar::os_isOk())
{
  if (laser.doProcessSimple(scan))
  {
    for (size_t i = 0; i < scan.points.size(); ++i)
    {
      const LaserPoint &p = scan.points.at(i);
      printf("r %.01f p %.01f\n",
        p.range * 1000.0f, p.intensity);
    }
    fflush(stdout);
  }
  else
  {
    fprintf(stderr, "[YDLIDAR] Failed to get lidar data\n");
    fflush(stderr);
  }
```

```
  }

  laser.turnOff();
  laser.disconnecting();

  return 0;
}
```

这里我们定位到while函数里边,

```
for (size_t i = 0; i < scan.points.size(); ++i)
      {
        const LaserPoint &p = scan.points.at(i);
        printf("r %.01f p %.01f\n",
          p.range * 1000.0f, p.intensity);
      }
```

这里就是获取到距离并且打印出来，实际的代码逻辑就是选择串口、初始化串口、雷达初始化、启动扫描、打印结果。具体的代码实现请自行解读。